Exp: 12(a) Echo client - server using TCP
17/9/2026

AIM:

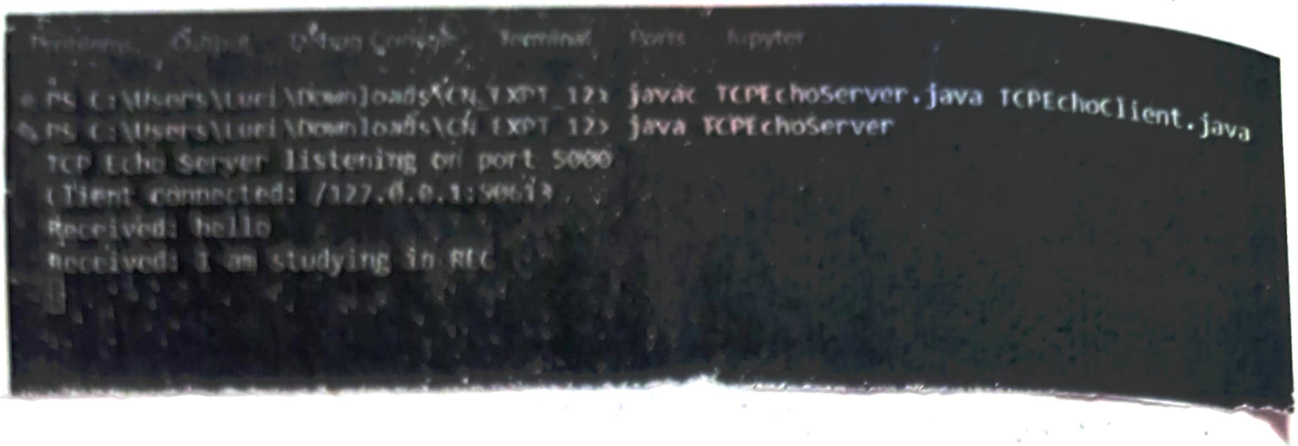
To implement an Echo client-server application using TCP Socket and verified communication b/w the Client and Server.

Procedure:

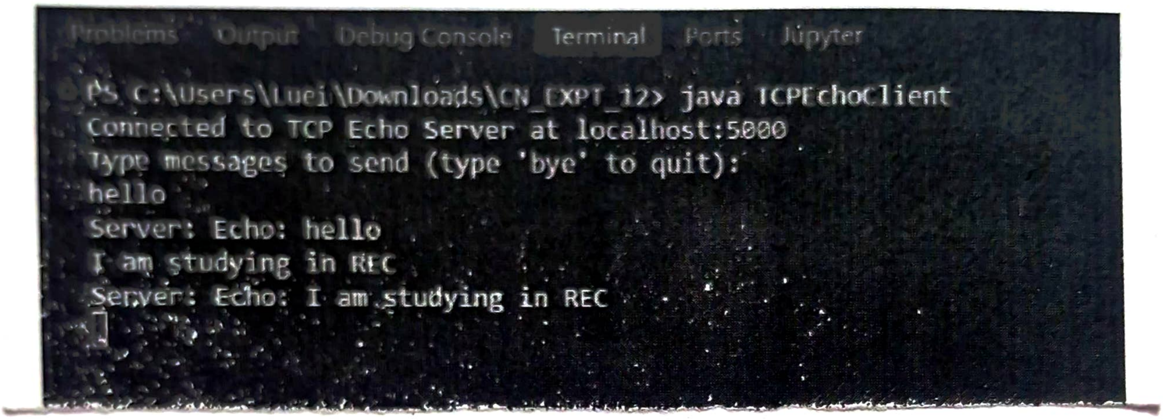
1. Create a TCP Echo Server using ServerSocket and Socket
2. Assign a port number to the Server.
3. Compile and execute the TCP Server Program.
4. Creat a TCP Echo client using Socket
5. Connect the Client to the Server using the Server address and port number.
6. Sends a message through the Client
7. The Server recives the message and Send it back to Client.
8. Display the echoed message at the Client
9. Enter bye to terminate the Communication.

Result!

Thus the TCP Echo Client - Server application was successfully implemented and the Client successfully received the echoed message from the Server.


Problems
Output
Debug Console
Terminal
Ports
Jupyter

```
PS C:\Users\Luei\Downloads\CN_EXPT_12> javac TCPEchoServer.java TCPEchoClient.java
PS C:\Users\Luei\Downloads\CN_EXPT_12> java TCPEchoServer
TCP Echo Server listening on port 5000
Client connected: /127.0.0.1:50613
Received: hello
Received: I am studying in REC
```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```
PS C:\Users\Luei\Downloads\CN_EXPT_12> java TCPEchoClient
Connected to TCP Echo Server at localhost:5000
Type messages to send (type 'bye' to quit):
hello
Server: Echo: hello
I am studying in REC
Server: Echo: I am studying in REC
```

Code:

```
~~Imp~~ Import java.io
    Import java.net;
 Public class chatserver{
Public class chatserver{
    Public static void;
  s = socket.socket()
    s.bind(("localhost", 5000))
     s.lit(1)
     s.lnt(1)
      c, addr = s.accept()
       msg = c.recv(1024)
       c.send(msg)
        c.close()
        s.close
```

TCP echo client

```
Import socket
s = socket.socket()
s.connect(("localhost", 500
s.send(b"Hello server")
Print(s.recv(1024).decode())
 s.close;
```